

EduFlow

Complete Performance Optimization & Full Responsive Design

Top-to-Bottom Fix Guide — [Frontend](#) · [Backend](#) · [Database](#)

Area	Problem	Fix	Impact
React Query	Stale data re-fetches	staleTime 5min + gcTime	−1.0s per nav
Double Fetch	useEffect + queryKey both fire	Remove manual refetch	−1.5s on filter
Similar Courses	Fetches all 193 for 3 cards	Server-side category filter	−0.8s
Learn Page	Fetches all, filters in JS	Direct /by-id/{id} endpoint	−0.6s
Search	API call on every keystroke	400ms debounce + minChars:2	−N calls
Backend Cache	No caching on GET endpoints	ETag + 304 Not Modified	−0.8s repeat
SQL Logging	TRACE logs 200 lines/request	show-sql: false → WARN	−0.3s dev
N+1 Queries	LAZY collections in loops	JOIN FETCH in repository	−variable
DB Indexes	Seq Scan on category/slug	4 indexes via migration	−variable
Skeleton UI	Blank page on first load	animate-pulse skeletons	Perceived perf
Navbar	No mobile menu	Hamburger + slide drawer	Mobile usable
Homepage Slider	Image overflow + float card	Responsive height + stack	Mobile fixed
Course Card	Hover popover on touch	matchMedia check on mount	Touch fixed
Filter Sidebar	Full-width blocks grid	Slide-in drawer on mobile	Mobile usable
Course Detail	pr-[360px] overflow mobile	Remove on mobile + stack	Layout fixed
Checkout	2-col broken on mobile	flex-col stack on mobile	UX fixed
Learn Page UI	Sidebar off-screen mobile	Collapsible below video	Mobile fixed
Images global	Overflow / no aspect ratio	max-w-full + aspect-[16/10]	All screens

SECTION 0 — What is Responsive Design? (Concept)

0.1 Core Idea

Responsive design means your website **automatically adjusts its layout, font sizes, spacing, and images** based on the screen width of the device. One codebase — perfect on all devices.

Device	Screen Width	Tailwind Prefix	What changes
Mobile	0 – 639px	default (no prefix)	Single column, hamburger nav, stacked layout
Small	640 – 767px	sm:	2-column grids, slightly larger text
Tablet	768 – 1023px	md:	Sidebar appears, 3-col grid, desktop nav
Laptop	1024 – 1279px	lg:	Full sidebar fixed, 4-col grid
Desktop	1280px+	xl:	Max-width container centered, wider spacing

0.2 How Tailwind Responsive Works

In Tailwind CSS, you write classes **mobile-first**. The default class applies to all screens. Prefixed classes override at that breakpoint and above.

```
// Mobile-first example:
className="text-sm md:text-base lg:text-lg"
// ^mobile ^tablet+ ^laptop+

// Grid example:
className="grid grid-cols-1 sm:grid-cols-2 md:grid-cols-3 lg:grid-cols-4"
// ^mobile ^640px+ ^768px+ ^1024px+

// Show/hide example:
className="hidden md:flex" // hidden on mobile, flex on tablet+
className="flex md:hidden" // visible on mobile, hidden on tablet+
```

0.3 Three Pillars of Responsiveness

1. Fluid Layouts	2. Responsive Images	3. Responsive Typography
Use % or grid/flex instead of fixed px widths	Images should never overflow 100% or contain full-size screenshots. Use <code>max-width: 100%; height: auto;</code>	Use <code>text-base</code> or <code>font-size: 1em;</code> as a base. Scale up with <code>text-lg</code> , <code>font-size: 1.2em;</code> or <code>font-size: 120%;</code>

SECTION 1 — Performance Optimization (Fix 2–3 Second Delays)

Every fix below targets a specific cause of the 2–3 second loading time. Implement in order — biggest impact first.

1.1 React Query Global Config

[MODIFY] frontend/src/components/QueryProvider.tsx

■ **Default staleTime is 0 — React Query re-fetches on EVERY component mount. Every page navigation = new API call.**

```
const queryClient = new QueryClient({
  defaultOptions: {
    queries: {
      staleTime: 5 * 60 * 1000, // 5 min — no re-fetch on navigation
      gcTime: 10 * 60 * 1000, // 10 min — keep data in memory
      refetchOnWindowFocus: false, // no re-fetch on tab switch
      refetchOnReconnect: false, // no burst fetch on reconnect
      retry: 1, // only retry once on failure
    },
  },
});
```

■ Impact: Navigating Home → Courses → Home no longer triggers 3 API calls. Data is served from cache instantly.

1.2 Fix Double-Fetch Bug — Courses Catalog Page

[FIX] frontend/src/app/courses/page.tsx

■ **CRITICAL BUG: useEffect calls refetch() manually when queryParam/categoryParam changes. But React Query ALREADY auto-refetches when queryKey changes. This causes 2 API calls per filter action — doubling the wait time.**

DELETE this block entirely:

```
// ■ REMOVE — causes double fetch
useEffect(() => {
  refetch();
}, [queryParam, categoryParam]);
```

This alone is sufficient — queryKey change auto-triggers refetch:

```
// ■ KEEP — this handles everything
const { data, isLoading } = useQuery({
  queryKey: ['courses', queryParam, categoryParam],
  queryFn: () => fetchCourses(queryParam, categoryParam),
});
```

■ Impact: Every filter/search action goes from 2 API calls to 1. Saves 1–1.5 seconds per interaction.

1.3 Course Detail — Stop Fetching All 193 Courses for Similar Section

[MODIFY] frontend/src/app/courses/[slug]/page.tsx

■ **Fetching all 193 courses over the network just to show 3 'similar courses' wastes 400ms+ bandwidth every time a user opens any course.**

```
// ■ WRONG – fetches 193 courses, filters 3 in browser
const { data: all } = useQuery({ queryKey: ['courses'] })
const similar = all?.filter(c => c.category === course.category).slice(0,3)
// ■ CORRECT – server returns only 3 matching courses
const { data: similar } = useQuery({
  queryKey: ['courses', 'similar', course.category, course.id],
  queryFn: () => api.get('/api/courses', {
    params: { category: course.category, limit: 3, exclude: course.id }
  }),
  enabled: !!course?.category,
});
```

■ Backend: Add 'limit' and 'exclude' query params to GET /api/courses in CourseController.

1.4 Learn Page — Fetch Course by ID Directly

[MODIFY] frontend/src/app/learn/[courseId]/page.tsx

■ Learn page currently fetches ALL courses via /api/courses then does array.find(id) in JavaScript. This loads the entire catalog just to get one course.

```
// ■ WRONG
const { data: all } = useQuery({ queryKey: ['courses'] })
const course = all?.find(c => c.id === courseId)
// ■ CORRECT – fetch the single course directly
const { data: course } = useQuery({
  queryKey: ['course', 'detail', courseId],
  queryFn: () => api.get(`/api/courses/by-id/${courseId}`),
  enabled: !!courseId,
});
```

Add this endpoint in CourseController.java:

```
@GetMapping("/api/courses/by-id/{id}")
public ResponseEntity<CourseDTO> getCourseById(@PathVariable UUID id) {
  return courseService.findById(id)
    .map(ResponseEntity::ok)
    .orElse(ResponseEntity.notFound().build());
}
```

1.5 Search Autocomplete — Debounce + Min Characters

[MODIFY] frontend/src/components/Navbar.tsx

```
// ■ 400ms debounce + skip single-char queries
const debouncedSearch = useDebounceCallback((val: string) => {
  if (val.trim().length < 2) {
    setAutocompleteResults([]);
    return;
  }
  fetchAutocomplete(val);
}, 400); // was 200ms
```

■ Impact: User typing 'react' fires 0 calls for 'r', then 1 call after 400ms of pause. Saves 4-5 wasted API calls per search.

1.6 Backend — ETag Caching (Safe for Dynamic Data)

[MODIFY] backend/.../CourseController.java

■ **Do NOT use Cache-Control: max-age=300.** If a course price changes or new course is added, users get stale data for 5 minutes. Use ETags instead — always accurate, still fast on repeat requests.

```
@GetMapping("/api/courses")
public ResponseEntity<List<CourseDTO>> getCourses(
    @RequestParam(required=false) String category,
    @RequestParam(required=false) Integer limit,
    WebRequest request) {
    List<CourseDTO> courses = courseService.findAll(category, limit);
    String etag = "'" + Integer.toHexString(courses.hashCode()) + "'";
    if (request.checkNotModified(etag)) return null; // returns 304
    return ResponseEntity.ok()
        .eTag(etag)
        .cacheControl(CacheControl.noCache().mustRevalidate())
        .body(courses);
}
```

■ Result: Second load of same course list returns HTTP 304 (no body) — near-instant response.

1.7 Disable SQL Logging in Dev (Huge I/O Killer)

[MODIFY] backend/src/main/resources/application-dev.yml

■ **TRACE-level Hibernate logging prints 50-200 lines per request to console/file.** With file-backed loggers this adds 200-500ms per request in dev.

```
spring:
  jpa:
    show-sql: false # was: true
  properties:
    hibernate:
      format_sql: false # was: true
      logging:
        level:
          org.hibernate.SQL: WARN # was: DEBUG/TRACE
          org.hibernate.type.descriptor.sql: WARN # stops param logging
          org.springframework.web: INFO
```

1.8 Fix N+1 Query Problem in JPA

[MODIFY] backend/.../CourseRepository.java

■ **N+1 problem: Loading 10 courses with LAZY lectures = 1 query for courses + 10 queries for lectures = 11 queries.** This is the #1 hidden cause of slow responses in Spring Boot.

How to detect N+1 (temporarily enable SQL log, check count):

```
# Temporarily in application-dev.yml to audit:
logging.level.org.hibernate.SQL: DEBUG
# Count the SELECT statements — if you see 10+ for one page load = N+1
```

Fix with JOIN FETCH:

```
// ■ PROBLEM — LAZY fetch causes N+1
@OneToMany(fetch = FetchType.LAZY)
private List<Lecture> lectures;

// ■ FIX — use JOIN FETCH in custom query
@Query("SELECT c FROM Course c LEFT JOIN FETCH c.lectures WHERE c.slug = :slug")
Optional<Course> findBySlugWithLectures(@Param("slug") String slug);

// ■ For catalog (no lectures needed — use projection)
@Query("SELECT c FROM Course c WHERE (:cat IS NULL OR c.category = :cat)")
Page<Course> findByCategory(@Param("cat") String cat, Pageable p);
```

1.9 Add Database Indexes

[ADD] backend/src/main/resources/db/migration/V3__add_indexes.sql

■ Without indexes, PostgreSQL does a full table scan (Seq Scan) on every query. On 193+ courses, this adds 200-800ms per request.

```
-- Run EXPLAIN ANALYZE first to confirm Seq Scan exists
EXPLAIN ANALYZE SELECT * FROM courses WHERE category = 'web-development';

-- Add these indexes:
CREATE INDEX IF NOT EXISTS idx_courses_category ON courses(category);
CREATE INDEX IF NOT EXISTS idx_courses_slug ON courses(slug);
CREATE INDEX IF NOT EXISTS idx_courses_instructor ON courses(instructor_id);
CREATE INDEX IF NOT EXISTS idx_courses_created_at ON courses(created_at DESC);
CREATE INDEX IF NOT EXISTS idx_enrollments_user ON enrollments(user_id);
CREATE INDEX IF NOT EXISTS idx_enrollments_course ON enrollments(course_id);
CREATE INDEX IF NOT EXISTS idx_lectures_course ON lectures(course_id);

-- After adding, verify with:
EXPLAIN ANALYZE SELECT * FROM courses WHERE category = 'web-development';
-- Should now show: Index Scan instead of Seq Scan
```

1.10 Skeleton Loaders — Perceived Performance (All Pages)

[ADD] frontend/src/components/skeletons/

■ Even after all fixes, first-load still shows a blank white page for 300-500ms. Skeleton loaders make the app feel instant.

```
// CourseCardSkeleton.tsx
export const CourseCardSkeleton = ({ count = 6 }) => (
  <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-4">
    {Array(count).fill(0).map((_, i) => (
      <div key={i} className="animate-pulse rounded-lg overflow-hidden">
        <div className="bg-gray-700 aspect-[16/10] w-full" />
        <div className="p-3 space-y-2">
          <div className="bg-gray-700 h-4 rounded w-full" />
          <div className="bg-gray-700 h-3 rounded w-3/4" />
          <div className="bg-gray-700 h-3 rounded w-1/2" />
        </div>
      </div>
    ))}
  </div>
);

// Usage in courses/page.tsx:
if (isLoading) return <CourseCardSkeleton count={12} />;
```

Page	Skeleton to Show	Count
Homepage	Hero skeleton + CourseCardSkeleton	8 cards

Courses	CourseCardSkeleton in grid	12 cards
Course Detail	Banner skeleton + sidebar skeleton	1 each
Learn Page	Video block skeleton + curriculum list	1 + 8
Checkout	Form skeleton + order summary skeleton	1 each

SECTION 2 — Full Responsive Design (Top to Bottom)

2.1 Global CSS Fixes (Apply First — Affects All Pages)

[MODIFY] frontend/src/app/globals.css

```
/* === GLOBAL RESPONSIVE RESETS === */
/* Prevent horizontal scroll on ALL screens */
*, ::before, ::after { box-sizing: border-box; }
body { overflow-x: hidden; margin: 0; }
/* All images responsive by default */
img { max-width: 100%; height: auto; display: block; }
/* Skeleton pulse animation */
@keyframes pulse {
  0%, 100% { opacity: 1; }
  50% { opacity: 0.4; }
}
.animate-pulse { animation: pulse 1.8s ease-in-out infinite; }
/* iOS safe area – content above home bar */
@supports (padding: env(safe-area-inset-bottom)) {
  .safe-bottom { padding-bottom: env(safe-area-inset-bottom); }
}
/* Responsive container */
.container { width: 100%; max-width: 1280px; margin: 0 auto;
padding-left: 1rem; padding-right: 1rem; }
@media (min-width: 768px) { .container { padding-left: 2rem; padding-right: 2rem; } }
@media (min-width: 1024px){ .container { padding-left: 3rem; padding-right: 3rem; } }
```

2.2 Navbar — Hamburger Menu + Mobile Search Overlay

[MODIFY] frontend/src/components/Navbar.tsx

Mobile layout changes:

- Show  **hamburger button** on mobile (hidden on md+)
- Slide-in drawer from left — contains Categories, My Learning, Wishlist, Login/Signup
- Search bar: **hidden on mobile** — replace with  icon that opens full-width search overlay
- Logo: **text-lg md:text-xl** — smaller on mobile
- Auth buttons: stacked vertically in the drawer on mobile


```
// Navbar.tsx – key structure
const [drawerOpen, setDrawerOpen] = useState(false);
const [searchOpen, setSearchOpen] = useState(false);
return (
<nav className="fixed top-0 left-0 right-0 z-50 bg-gray-900 h-14">
<div className="flex items-center justify-between h-full px-4">
  {/* Hamburger – mobile only */}
  <button className="flex md:hidden" onClick={() => setDrawerOpen(true)}>☰</button>
  {/* Logo */}
  <span className="text-lg md:text-xl font-bold text-purple-500">EduFlow</span>
  {/* Search – desktop only */}
  <div className="hidden md:flex flex-1 mx-6">
    <SearchBar />
  </div>
  {/* Search icon – mobile only */}
  <button className="flex md:hidden" onClick={() => setSearchOpen(true)}>🔍</button>
  {/* Desktop nav links */}
  <div className="hidden md:flex items-center gap-4">
    <NavLinks /> <AuthButtons />
  </div>
</div>
</div>
  {/* Mobile drawer */}
  {drawerOpen && (
    <div className="fixed inset-0 z-50">
      <div className="absolute inset-0 bg-black/60" onClick={() => setDrawerOpen(false)} />
      <div className="absolute left-0 top-0 bottom-0 w-72 bg-gray-900 p-6 flex flex-col gap-4">
        <NavLinks vertical /> <AuthButtons vertical />
      </div>
    </div>
  )}
  {/* Mobile search overlay */}
  {searchOpen && (
    <div className="fixed inset-0 z-50 bg-gray-900 p-4">
      <SearchBar autoFocus onClose={() => setSearchOpen(false)} />
    </div>
  )}
</nav>
);
```

2.3 Homepage — Hero Slider, Dynamic Image, Course Grids

[MODIFY] frontend/src/app/page.tsx

Hero Slider — 3 specific issues to fix:

Issue	Cause	Fix
Image overflows on mobile	Fixed height 360px on container	Use min-h-[200px] md:min-h-[360px] + object-cover
Slider doesn't resize	Fixed pixel width on slides	Use w-full on each slide, overflow-hidden on container
Floating card overlaps img	absolute positioned over image	hidden on mobile, show below image as normal block

```
// Hero Slider — responsive fix
<section className="relative overflow-hidden">
  {/* Slider container — responsive height */}
  <div className="relative min-h-[200px] sm:min-h-[280px] md:min-h-[360px]">
    {/* Slide image — always full width, correct height */}
    <img
      src={slide.image}
      alt={slide.title}
      className="w-full h-full object-cover absolute inset-0"
    />
    {/* Text overlay — responsive padding */}
    <div className="relative z-10 p-4 md:p-8 lg:p-16">
      <h1 className="text-lg sm:text-2xl md:text-4xl font-bold text-white">
        {slide.title}
      </h1>
      <p className="text-sm md:text-base text-gray-200 mt-2">{slide.subtitle}</p>
    </div>
    {/* Floating promo card — DESKTOP ONLY */}
    <div className="hidden md:block absolute right-8 top-8 w-64 bg-white rounded-lg p-4">
      <PromoCard />
    </div>
  </div>
  {/* Floating card on MOBILE — below slider as normal block */}
  <div className="md:hidden p-4">
    <PromoCard />
  </div>
</section>
```

Course Grids — consistent breakpoints across all sections:

```
// Use this class on ALL course grids on homepage
className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-4 md:gap-6"
// Company/partner logo marquee
className="flex gap-4 md:gap-10 items-center"
// Logo images:
className="h-5 md:h-7 w-auto object-contain opacity-70"
```

2.4 Course Card — Full Responsive + Fix Touch Hover

[MODIFY] frontend/src/components/CourseCard.tsx

Hover Popover Fix — check once on mount only:

```
const [isHoverDevice, setIsHoverDevice] = useState(false);
useEffect(() => {
  // Run ONCE on mount — matchMedia is expensive, don't run on every hover
  setIsHoverDevice(window.matchMedia('(hover: hover)').matches);
}, []); // empty deps = mount only
// Attach hover events conditionally
<div
  onMouseEnter={isHoverDevice ? () => setShowPopover(true) : undefined}
  onMouseLeave={isHoverDevice ? () => setShowPopover(false) : undefined}
>
```

Card responsive layout:

```

<div className="rounded-lg overflow-hidden bg-gray-800 hover:shadow-xl transition-shadow">
  {/* Thumbnail — fixed aspect ratio on ALL screens */}
  <div className="aspect-[16/10] overflow-hidden">
    <img
      src={course.thumbnail}
      alt={course.title}
      className="w-full h-full object-cover hover:scale-105 transition-transform duration-300"
    />
  </div>
  {/* Card body */}
  <div className="p-2 md:p-3">
    <h3 className="text-sm md:text-base font-semibold line-clamp-2 text-white">
      {course.title}
    </h3>
    <p className="text-xs text-gray-400 mt-1">{course.instructor}</p>
    <div className="flex items-center gap-1 mt-1">
      <StarRating rating={course.rating} size="sm" />
      <span className="text-xs text-gray-400">({course.reviewCount})</span>
    </div>
    <p className="text-sm md:text-base font-bold text-white mt-1">
      ■{course.price}
    </p>
  </div>
</div>

```

2.5 Courses Catalog — Filter Sidebar as Mobile Drawer

[MODIFY] frontend/src/app/courses/page.tsx

■ On mobile, the filter sidebar takes full width and completely hides the course grid. Must become a slide-in drawer.

```

const [filterOpen, setFilterOpen] = useState(false);
const [activeFilterCount, setActiveFilterCount] = useState(0);
return (
<div className="flex flex-col md:flex-row gap-6">
  {/* Mobile filter toggle button */}
  <div className="flex md:hidden items-center justify-between px-4 py-2 border-b border-gray-700">
    <span className="text-sm text-gray-300">{totalCourses} results</span>
    <button
      className="flex items-center gap-2 text-sm font-medium text-white"
      onClick={() => setFilterOpen(true)}
    >
      Filter {activeFilterCount > 0 && (
        <span className="bg-purple-600 text-white text-xs rounded-full px-1.5">{activeFilterCount}</span>
      )}
    </button>
  </div>
  {/* Sidebar — desktop: inline, mobile: drawer */}
  <aside className="hidden md:block w-56 flex-shrink-0">
    <FilterPanel />
  </aside>
  {/* Mobile filter drawer */}
  {filterOpen && (
    <div className="fixed inset-0 z-50 md:hidden">
      <div className="absolute inset-0 bg-black/60" onClick={() => setFilterOpen(false)} />
      <div className="absolute right-0 top-0 bottom-0 w-80 bg-gray-900 overflow-y-auto p-4">
        <div className="flex justify-between mb-4">
          <h3 className="font-semibold text-white">Filters</h3>
          <button onClick={() => setFilterOpen(false)}>X</button>
        </div>
        <FilterPanel />
      </div>
    </div>
  )}
  {/* Course grid */}
  <main className="flex-1">
    <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-4">
      {courses.map(c => <CourseCard key={c.id} course={c} />)}
    </div>
  </main>
</div>
);

```

2.6 Course Detail Page — Full Mobile Layout

[MODIFY] frontend/src/app/courses/[slug]/page.tsx

Problem 1 — Header banner overflow fix:

```

// ■ WRONG — md:pr-[360px] applied on mobile = text pushed off screen
<div className="px-4 py-8 md:pr-[360px]">

// ■ CORRECT — padding only on md+ where sidebar exists
<div className="px-4 py-8 md:pl-8 md:pr-[380px]">

```

Problem 2 — Buy sidebar mobile placement:

```
// Desktop sidebar — fixed right, hidden on mobile
<aside className="hidden md:block fixed right-6 top-20 w-[320px]">
  <BuySidebarCard course={course} />
</aside>
// Mobile: full-width, below description (NOT above — let user read first)
<div className="md:hidden w-full mt-6 px-4">
  <BuySidebarCard course={course} fullWidth />
</div>
```

Problem 3 — Similar courses + buttons:

```
// Similar courses grid
className="grid grid-cols-1 sm:grid-cols-2 md:grid-cols-3 gap-4"
// All CTA buttons — full width on mobile
className="w-full md:w-auto px-6 py-3 bg-purple-600 text-white rounded-lg font-semibold"
```

2.7 Buy Sidebar Card — Responsive Internals

[MODIFY] frontend/src/components/BuySidebarCard.tsx

```
// BuySidebarCard.tsx
const BuySidebarCard = ({ course, fullWidth = false }) => (
  <div className={`bg-white rounded-lg shadow-xl overflow-hidden
    ${fullWidth ? 'w-full' : 'w-[320px]'}`}>
    {/* Course thumbnail preview */}
    <div className="aspect-video overflow-hidden">
      <img src={course.thumbnail} className="w-full h-full object-cover" />
    </div>
    <div className="p-4">
      <p className="text-2xl font-bold text-gray-900">■{course.price}</p>
      <button className="w-full mt-3 py-3 bg-purple-600 text-white font-semibold rounded-lg">
        Buy Now
      </button>
      <button className="w-full mt-2 py-3 border border-gray-300 font-semibold rounded-lg">
        Add to Cart
      </button>
    </div>
  </div>
);
```

2.8 Checkout / Payment Page — Mobile Stack Layout

[MODIFY] frontend/src/app/checkout/page.tsx

```
// Checkout page layout
<main className="min-h-screen bg-gray-50 py-6">
  <div className="container">
    <h1 className="text-xl md:text-2xl font-bold text-gray-900 mb-6">Checkout</h1>
    {/* Stack on mobile, side-by-side on tablet+ */}
    <div className="flex flex-col md:flex-row gap-6">
      {/* Form – full width on mobile */}
      <section className="flex-1 space-y-4">
        <div className="bg-white rounded-lg p-4 md:p-6">
          <h2 className="font-semibold mb-4">Payment Details</h2>
          {/* All inputs full width */}
          <input className="w-full border rounded-lg px-3 py-2 text-sm" />
          {/* Card number + expiry row */}
          <div className="flex flex-col sm:flex-row gap-3 mt-3">
            <input className="flex-1 border rounded-lg px-3 py-2 text-sm" placeholder="Card Number" />
            <input className="w-full sm:w-28 border rounded-lg px-3 py-2 text-sm" placeholder="MM/YY" />
            <input className="w-full sm:w-20 border rounded-lg px-3 py-2 text-sm" placeholder="CVV" />
          </div>
        </div>
      </section>
      {/* Order summary */}
      <aside className="w-full md:w-72 lg:w-80">
        <div className="bg-white rounded-lg p-4 md:p-6 sticky top-20">
          <OrderSummary />
        </div>
      </aside>
    </div>
  </div>
</main>
```

2.9 Learn Page — Video Player + Curriculum Sidebar

[MODIFY] frontend/src/app/learn/[courseId]/page.tsx

```

const [curriculumOpen, setCurriculumOpen] = useState(false);
<div className="flex flex-col lg:flex-row h-screen overflow-hidden">
  /* Main content */
  <main className="flex-1 overflow-y-auto">
    /* Video player – always full width, proper aspect ratio */
    <div className="w-full bg-black">
      <div className="aspect-video w-full">
        <VideoPlayer src={currentLecture.videoUrl} />
      </div>
    </div>

    /* Mobile curriculum toggle */
    <div className="lg:hidden border-t border-gray-700">
      <button
        className="w-full flex justify-between items-center px-4 py-3 text-sm font-semibold"
        onClick={() => setCurriculumOpen(!curriculumOpen)}
      >
        <span>Course Content</span>
        <span>{curriculumOpen ? '▲' : '▼'}</span>
      </button>
      {curriculumOpen && <CurriculumList lectures={lectures} />}
    </div>

    /* Lecture info below video */
    <div className="p-4 md:p-6">
      <h1 className="text-base md:text-xl font-bold">{currentLecture.title}</h1>
    </div>
  </main>

  /* Desktop curriculum sidebar */
  <aside className="hidden lg:block w-[360px] border-l border-gray-700 overflow-y-auto">
    <CurriculumList lectures={lectures} />
  </aside>
</div>

```

2.10 Footer

[MODIFY] frontend/src/components/Footer.tsx

- Already **grid-cols-2 md:grid-cols-4** — grid is fine
- Language dropdown: add **max-w-[180px]** — prevents overflow on 320px screens
- Add **safe-bottom** class to footer — prevents content hiding behind iOS home bar
- Social icons: **w-8 h-8 md:w-6 md:h-6** — larger tap target on mobile (min 44px recommended)

```

<footer className="bg-gray-900 pt-10 pb-6 safe-bottom">
  <div className="container">
    <div className="grid grid-cols-2 md:grid-cols-4 gap-6 md:gap-8">
      /* footer columns */
    </div>

    <div className="mt-8 border-t border-gray-700 pt-4 flex flex-col sm:flex-row
      justify-between items-center gap-3">
      <p className="text-xs text-gray-500">© 2024 EduFlow. All rights reserved.</p>
      <select className="max-w-[180px] text-xs border border-gray-600
        bg-transparent text-gray-300 rounded px-2 py-1">
        <option>English</option>
      </select>
    </div>
  </div>
</footer>

```

SECTION 3 — Image Responsiveness (Full Guide)

3.1 Image Rules — Apply Everywhere

Image Type	Problem	Fix Classes
Hero/Banner	Fixed height cuts on mobile	w-full h-full object-cover absolute inset-0
Course Card Thumb	Squishes/stretches on resize	aspect-[16/10] w-full object-cover
Course Detail Banner	Overflows small screens	w-full max-h-[300px] md:max-h-[400px] object-cover
Sidebar Card Thumb	Fixed width breaks mobile	w-full aspect-video object-cover
Instructor Avatar	Too large on mobile	w-10 h-10 md:w-14 md:h-14 rounded-full object-cover
Category Icons	Wrong size on mobile	w-8 h-8 md:w-10 md:h-10 object-contain
Logo images	Overflow or too small	h-5 md:h-7 w-auto object-contain
Learn page thumbnail	Wrong aspect on mobile	aspect-video w-full object-cover

3.2 Next.js Image Component — Use Properly

```
// ■ Always use next/image instead of <img> for:
// - automatic WebP conversion
// - lazy loading built in
// - prevents layout shift (CLS)
import Image from 'next/image';
// Course card thumbnail:
<div className="relative aspect-[16/10] w-full overflow-hidden rounded-t-lg">
  <Image
    src={course.thumbnail}
    alt={course.title}
    fill
    className="object-cover hover:scale-105 transition-transform duration-300"
    sizes="(max-width: 640px) 100vw, (max-width: 1024px) 50vw, 25vw"
  />
</div>
// Hero image (full width):
<div className="relative w-full min-h-[200px] md:min-h-[360px]">
  <Image
    src={slide.image}
    alt={slide.title}
    fill
    priority // loads immediately - above the fold
    className="object-cover"
    sizes="100vw"
  />
</div>
```


SECTION 4 — Full Verification Checklist

4.1 Performance Checks

#	Test	Expected Result
1	Open Network → navigate Home ↔ Courses 3x	No duplicate XHR for same queryKey
2	Filter courses by category, watch Network tab	Only 1 API call per filter (no double-fetch)
3	Open Course Detail → check 'similar' API call	Fetches max 3 courses, not 193
4	Open Learn page → check API call in Network	Calls /by-id/{id} not /api/courses
5	Type 'r' in search bar → watch XHR	Zero calls for single char
6	Type 'react' → watch XHR	1 call fires 400ms after last keystroke
7	GET /api/courses → check response headers	Second call returns HTTP 304, no body
8	Check backend terminal logs during browsing	No SQL lines printed (show-sql: false)
9	EXPLAIN ANALYZE on PostgreSQL course queries	Index Scan (not Seq Scan)
10	First load of Courses page	Skeleton cards visible within 100ms
11	Navigate between pages while skeleton active	Content appears progressively, no blank flash

4.2 Responsive Checks — By Viewport

Viewport	Test	Expected
375px	Navbar	Hamburger  visible, search bar hidden
375px	Hamburger tap	Slide drawer opens with all nav links
375px	Search icon tap	Full-screen search overlay opens
375px	Homepage hero slider	Image fills width, no overflow, promo card below
375px	Homepage course grid	1 column, cards full width
375px	Courses page	Filter button visible, course grid 1-col
375px	Courses filter button tap	Slide-in filter drawer from right
375px	Course Detail banner	No horizontal scroll, text readable
375px	Course Detail buy card	Full width, below description content
375px	Checkout	Single column, full-width inputs, no overflow
375px	Learn page	Video full-width 16:9, curriculum collapsible below
375px	Any Course Card	Tap card — popover does NOT appear
375px	Footer	Language dropdown not overflowing, safe area padding
375px	Any page	No horizontal scrollbar
768px	Navbar	Desktop links visible, hamburger hidden
768px	Courses	2-col grid, sidebar visible inline

768px	Homepage	2-col course grid, hero at medium height
1024px	Course Detail	Buy sidebar fixed right, desktop banner padding
1024px	Learn page	Curriculum sidebar fixed right 360px
1024px	Courses	3-col grid, full filter sidebar
1440px	All pages	4-col course grid, max-width container centered

4.3 Image Checks

Image	375px mobile	All screens
Hero slider image	Fills width, correct height, no crop	object-cover, no distortion
Course card thumbnail	16:10 ratio maintained	Never stretches/squishes
Course detail banner	Full width, max-height respected	Consistent across breakpoints
Buy sidebar thumbnail	Full width on mobile	aspect-video maintained
Instructor avatar	Visible, not too large	Circular, correct size per bp
Category/company logos	Not overflowing	h-5 md:h-7, object-contain

SECTION 5 — Implementation Order (Priority)

Priority	Task	Time Est.	Impact
1 — FIRST	Remove double-fetch useEffect	5 min	−1.5s immediately
2	globals.css resets	10 min	Prevents overflow bugs on all pages
3	show-sql: false in dev.yml	2 min	Faster dev environment
4	Skeleton loaders	2 hours	Perceived performance on all pages
5	QueryProvider staleTime	5 min	Stops re-fetches on navigation
6	Hero slider responsive fix	1 hour	Homepage looks correct on mobile
7	Navbar hamburger + drawer	3 hours	Mobile usable navigation
8	Course Card hover fix + layout	1 hour	Touch works, card looks right
9	Filter sidebar drawer	2 hours	Courses page mobile usable
10	Course Detail layout	1 hour	Detail page mobile fixed
11	Learn page mobile	1 hour	Video + curriculum mobile fixed
12	Checkout mobile	45 min	Payment works on mobile
13	Backend ETag caching	1 hour	Repeat loads instant
14	N+1 fix + DB indexes	2 hours	Database speed
15	Similar courses server filter	30 min	−0.8s on course detail
16	Learn page direct endpoint	30 min	−0.6s on learn page

17	Next.js Image components	2 hours	WebP, lazy load, no CLS
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